

Clustering

Agenda

- Introduction to Clustering
- Connectivity vs Centroid based Clustering
- K-Means Clustering
- Optimising the Number of Clusters

Let's begin the discussion by answering a few questions

Clustering Quiz

Which statement correctly differentiates connectivity-based and centroid-based clustering?

A

Connectivity-based minimises centroid distances; centroid-based uses neighbouring points.

B

Connectivity-based clusters by point connectivity; centroid-based clusters around centroids.

C

Connectivity-based needs predefined centres; centroid-based does not.

D

Connectivity-based relies on centroids; centroid-based relies on links between points.

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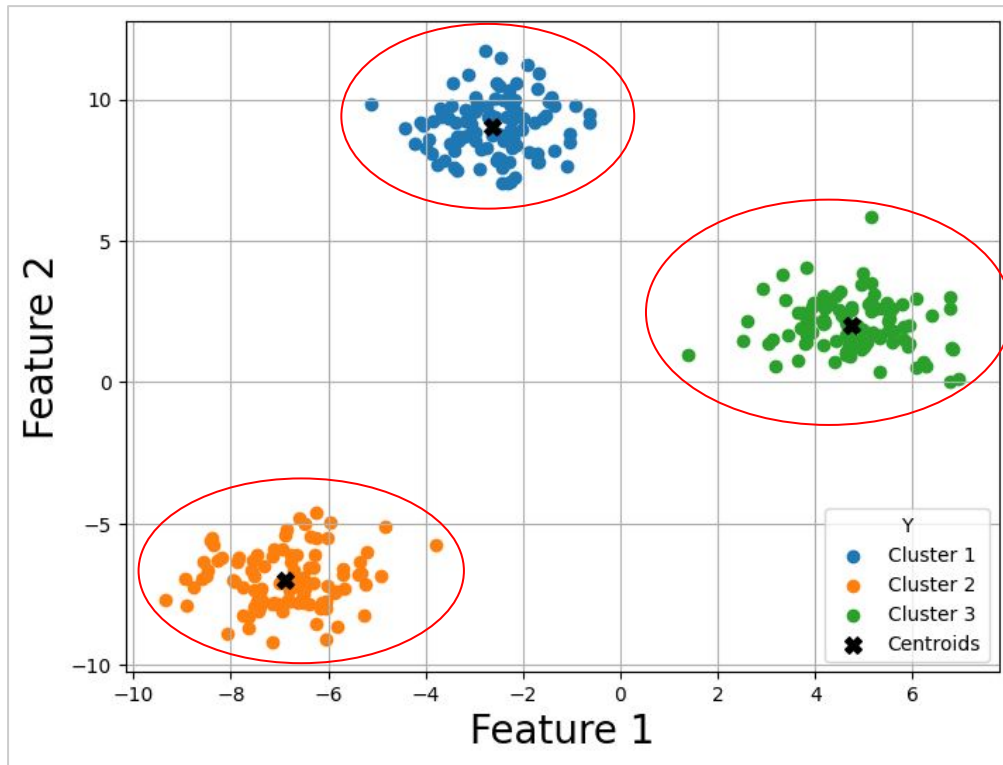
C

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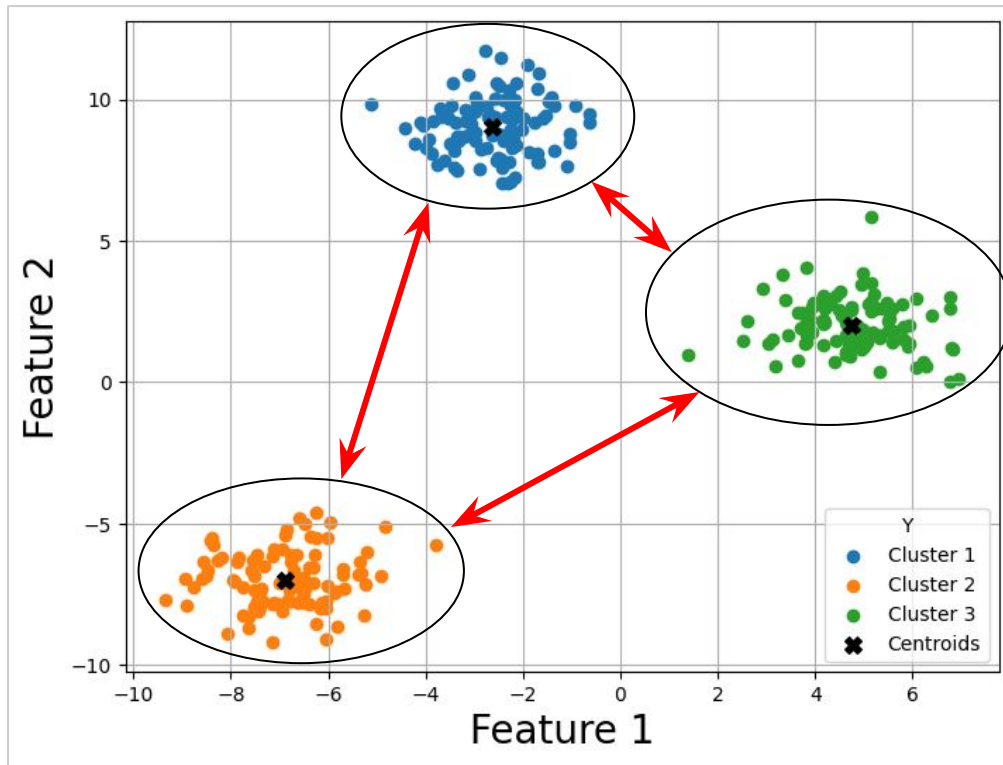
Connectivity-based relies on centroids; centroid-based relies on links between points.

The Goal of Clustering



Within each cluster,
the **variance is**
minimal

The Goal of Clustering



Between each cluster, the variance is maximized

Connectivity vs Centroid based Clustering

Connectivity-based Clustering

Group pairs of points based on how close they are.



Merge pairs to form bigger groups.



Keep repeating till we reach a desired number of groups.

Centroid-based Clustering

Assume a certain number of groups.



We can define a 'representative' for each group.



The points closest to these representatives can be grouped with them.

Clustering Quiz

In the K-Means clustering algorithm, what does the term "means" refer to?

A

The median of the data points in each cluster

B

The average distance between data points in a cluster

C

The average (mean) position of all data points assigned to a cluster

D

The total number of clusters formed in the dataset.

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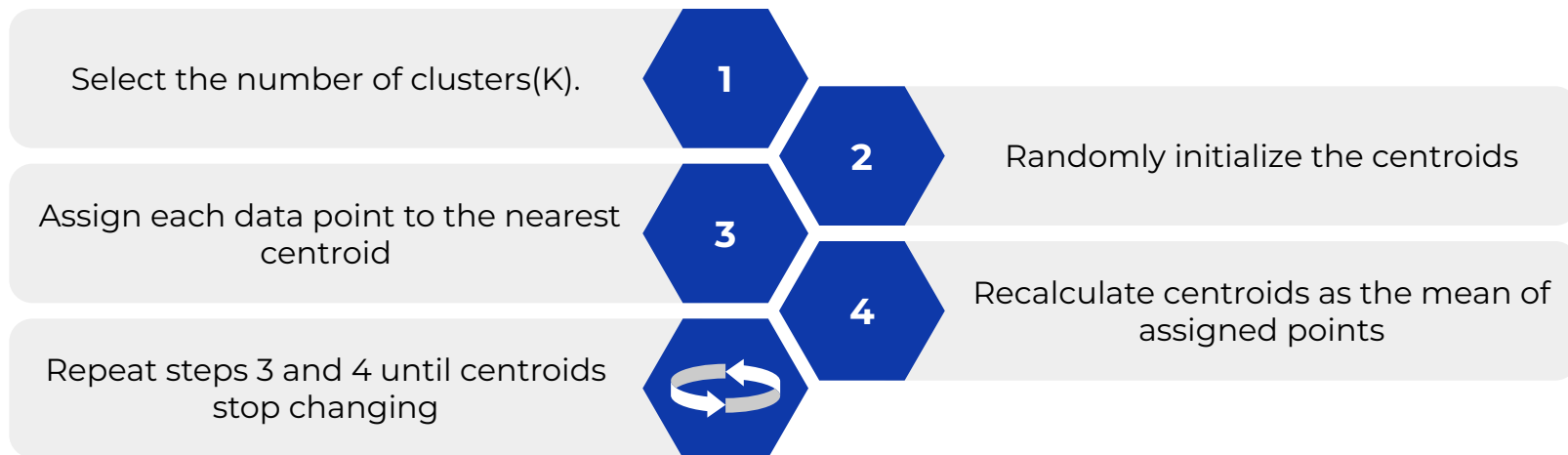
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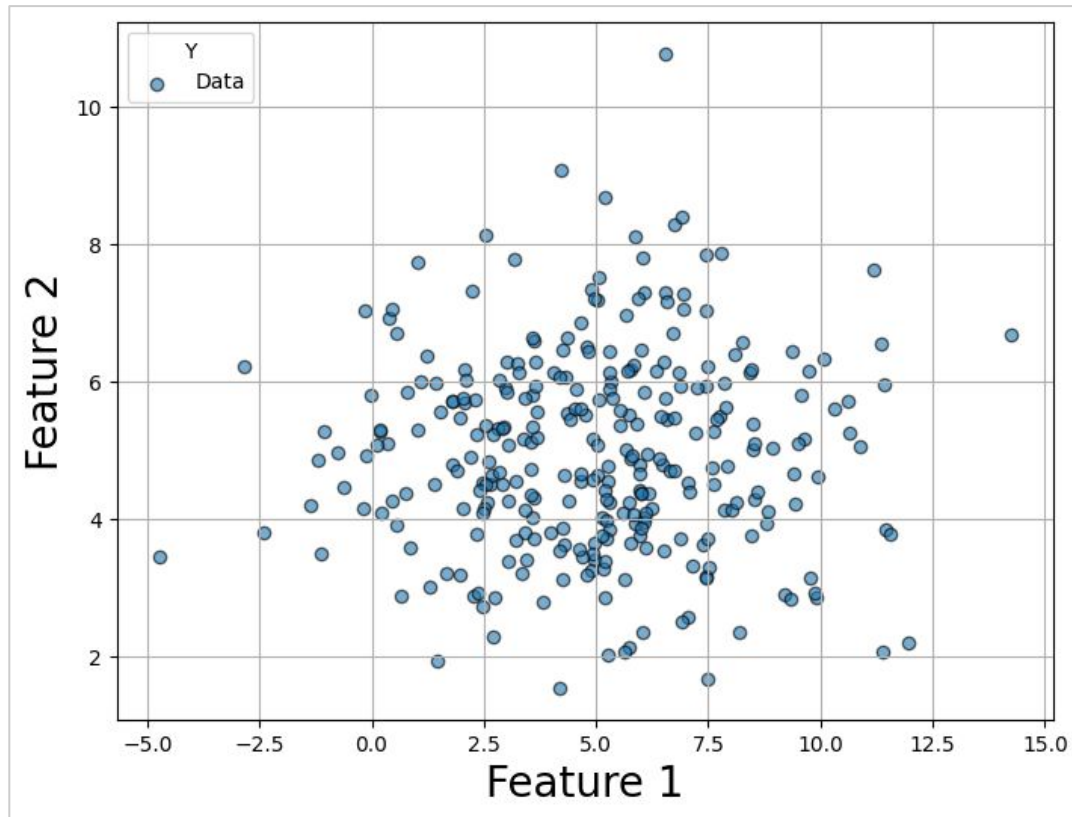
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K-Means Clustering

K-Means is an unsupervised clustering algorithm that groups similar data points into K clusters.



K-Means Clustering

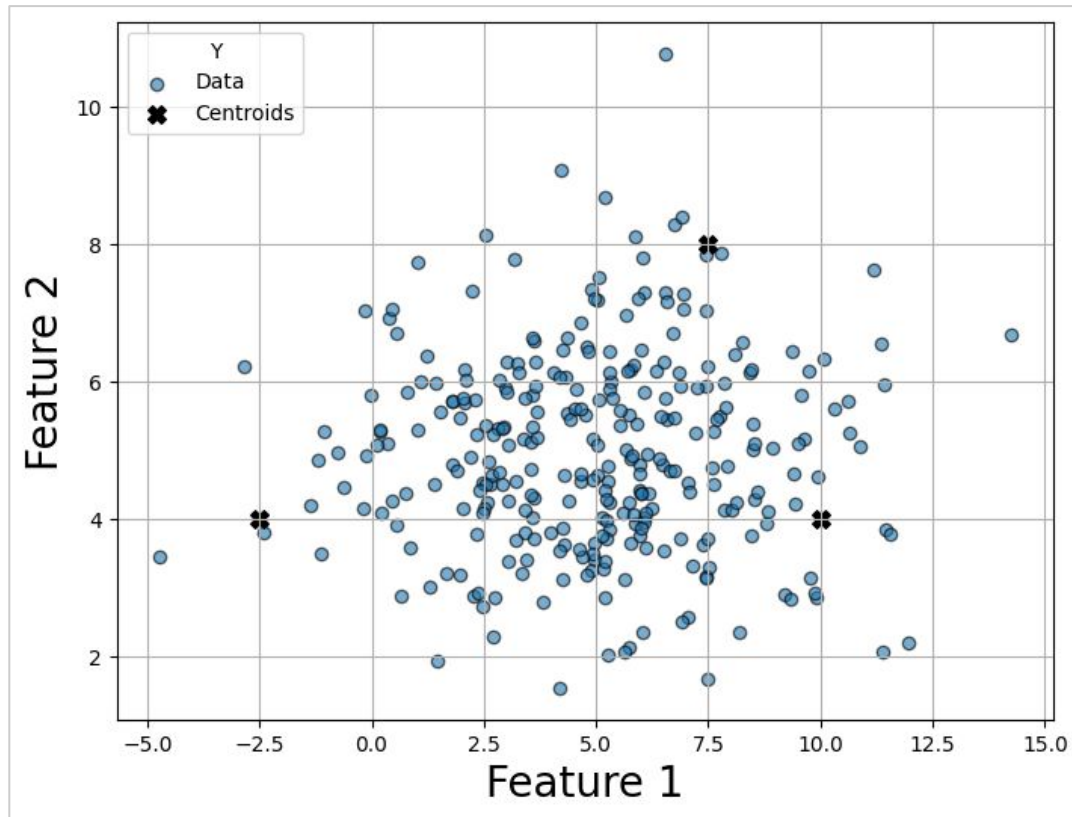


1

Select the number of clusters

Take $K = 3$

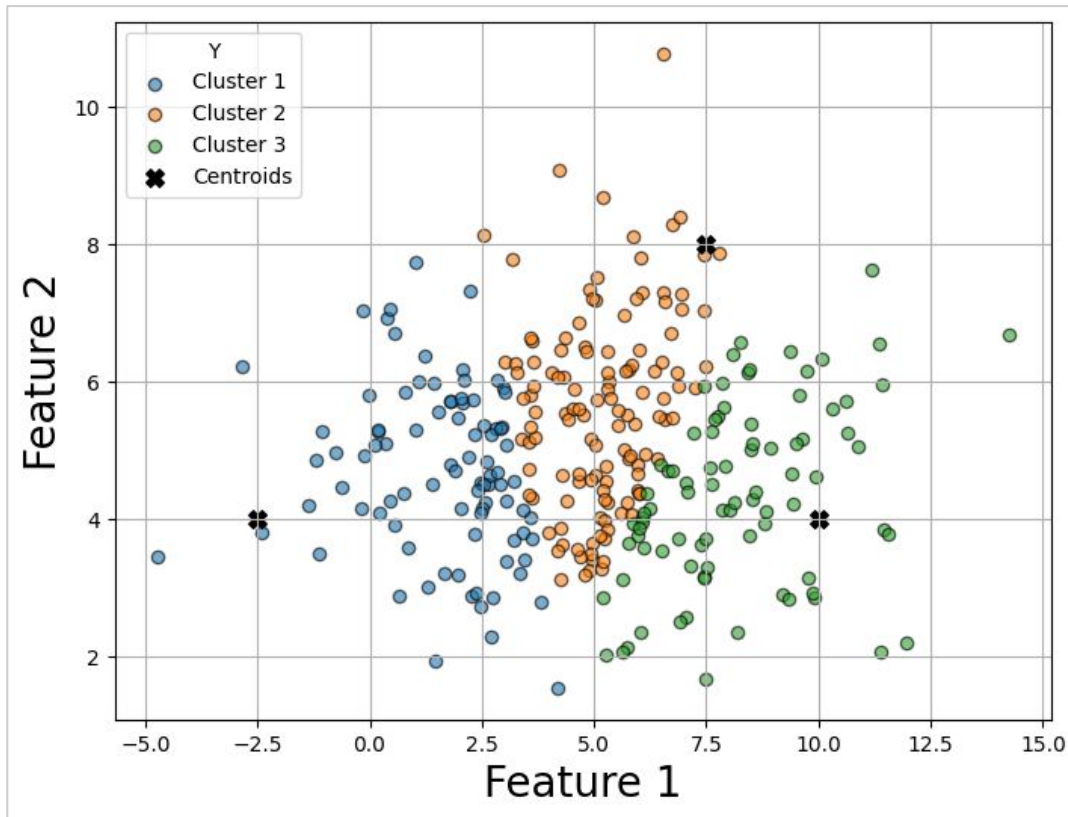
K-Means Clustering



2

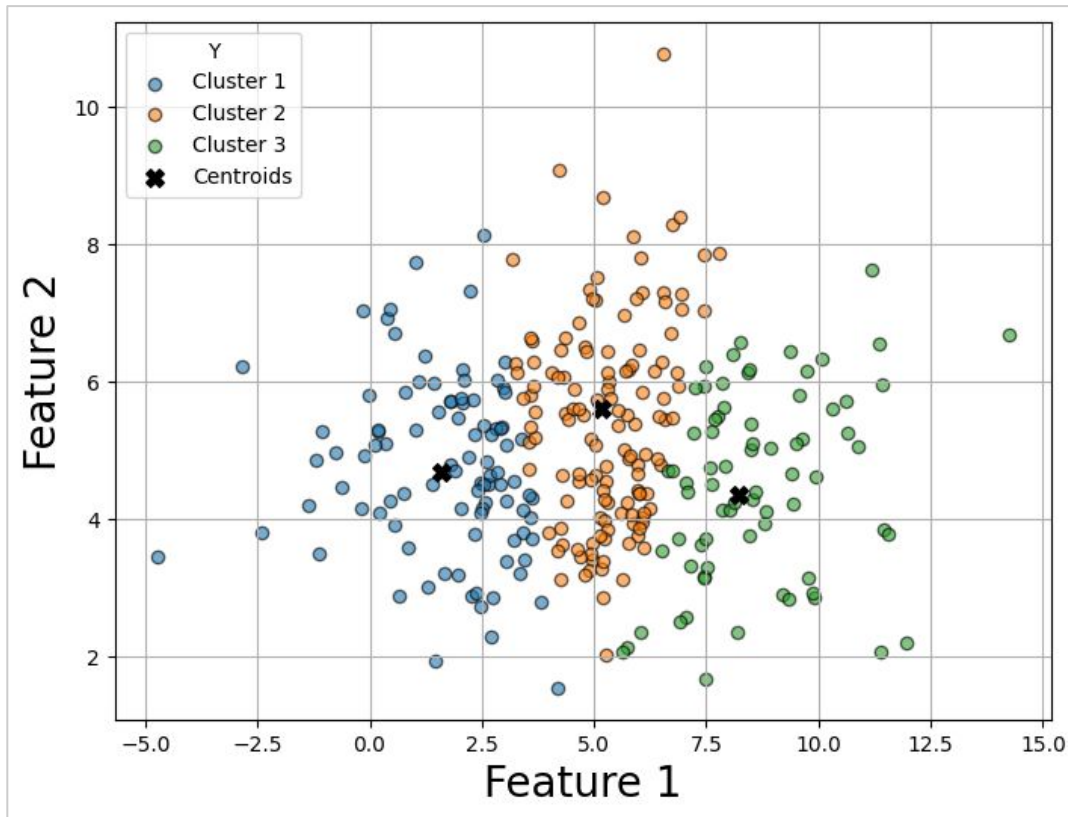
Randomly initialize the centroids

K-Means Clustering

**3**

Assign each data point to the nearest centroid

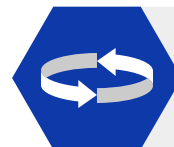
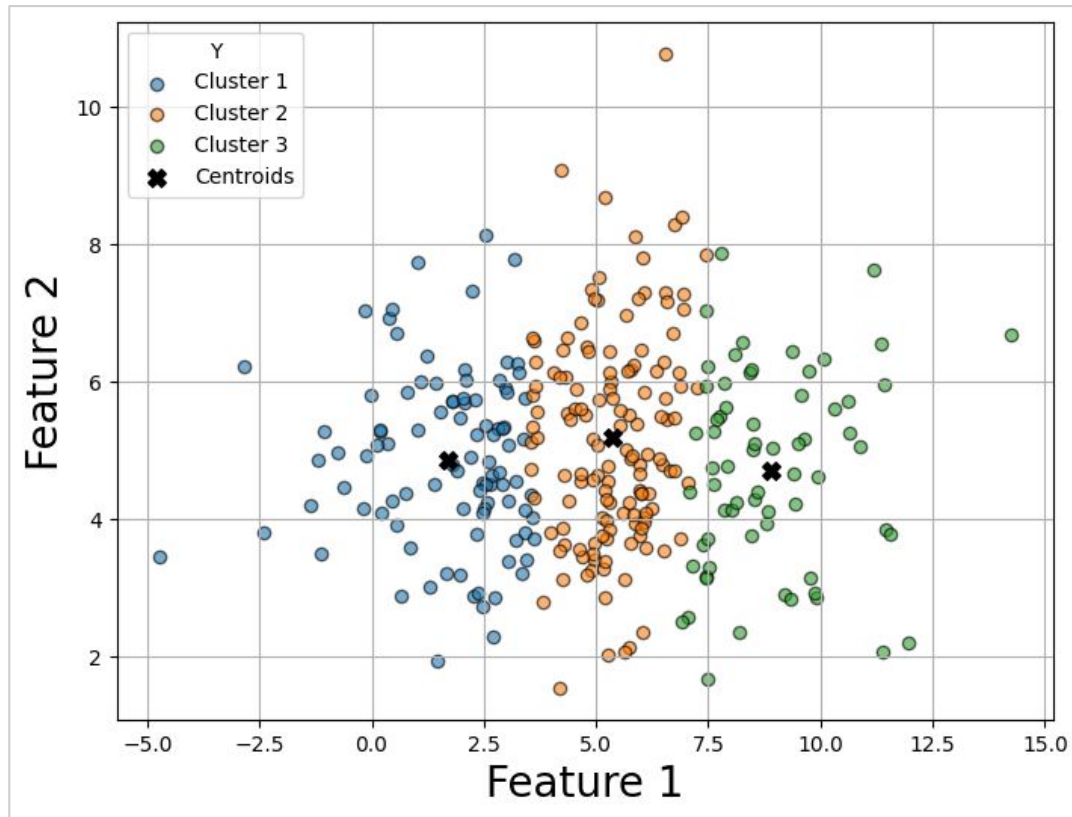
K-Means Clustering



4

Recalculate centroids as the mean of assigned points

K-Means Clustering



Repeat steps 3 and 4
until centroids stop
changing

Clustering Quiz

Which issue is a common limitation of standard K-Means clustering?

A

It cannot handle large datasets efficiently.

B

It forms clusters based on connectivity, not centroids.

C

Random initial centroids can lead to poor or inconsistent clustering.

D

It automatically chooses the optimal number of clusters without user input.

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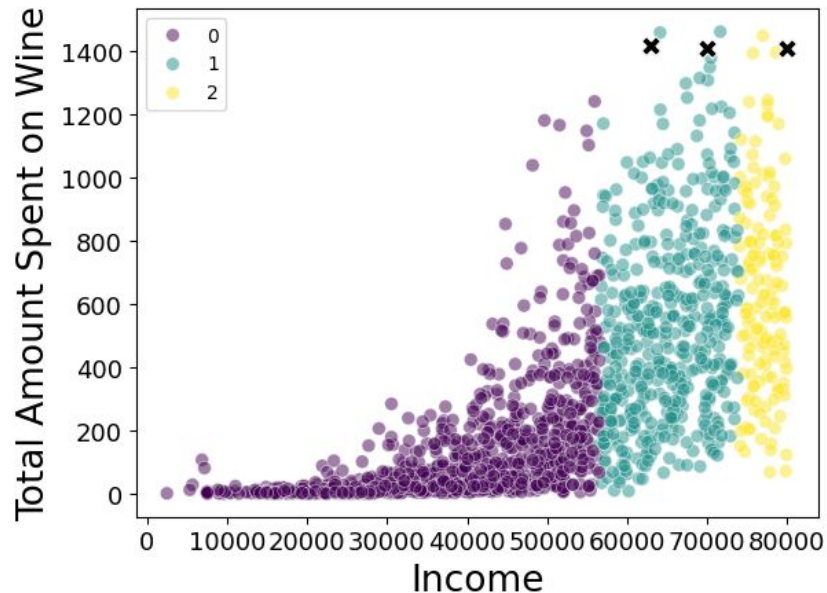
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Cluster Initialization



- Need to initialize the centroids well at the start to get better clusters.
- Poor initialization can lead to suboptimal clusters.

Use a technique called **K-Means++** for **initialization**.

- The first centroid is randomly chosen from the data points.
- For each data point, compute its distance to the nearest centroid already chosen.
- The next centroids are selected such that the data points farther away from already chosen centroids are more likely to be selected as the next centroid.
- Repeat until all K centroids are.

Clustering Quiz

In the Elbow Method, what does the "elbow" represent in the plot of WCSS vs. the number of clusters?

A

The point where the WCSS starts increasing significantly

B

The point where the number of clusters reaches its maximum

C

The point where the decrease in WCSS slows down, indicating the optimal number of clusters

D

The point where the clusters become perfectly separated with zero WCSS.

Clustering Quiz

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The Elbow Method with WCSS

1

Compute WCSS for different number of clusters (K)

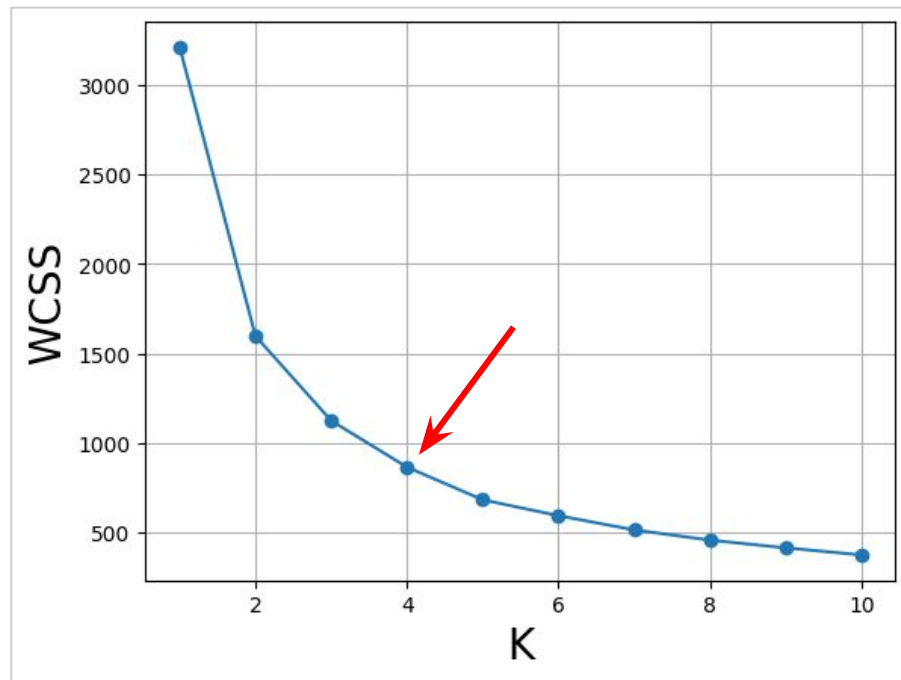
2

Plot WCSS vs. K for the different values of K

3

Observe a point where the rate of decrease of WCSS slows down sharply - like an 'elbow'

This point generally yields the **optimal number of clusters** for the data.



Similar to how Linear Regression minimizes the sum of squared errors to fit a line, K-Means minimizes the variance (Within-Cluster Sum of Squares) to fit clusters

Clustering Quiz

A negative Silhouette Score indicates that:

A

The clustering is perfect

B

The clustering is bad, the data points are likely assigned to the wrong clusters.

C

The clusters are compact but not well separated

D

The clustering quality cannot be evaluated using this score.

Clustering Quiz

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What is **silhouette score**?

- Let's choose one cluster (say K) and a point say (i) within it.

Similarity within a cluster

Find the average distance from the point to all the other points in the cluster it belongs to.

We denote this by $a(i)$.

Dissimilarity across clusters

Find the average distance between the point and all other points of the nearest cluster that the point is not a part of.

We denote this by $b(i)$.

Silhouette Score

- We would want to minimize $a(i)$ and maximize $b(i)$ – low intra-cluster distance and high inter-cluster distance.

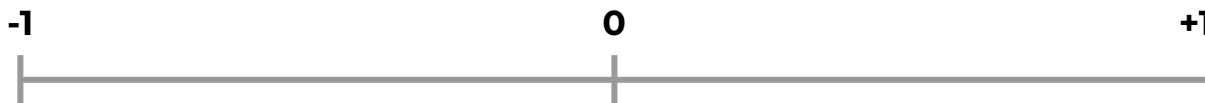
$$s(i) = \frac{b(i) - a(i)}{\max \{a(i), b(i)\}}$$

This is called the **silhouette score**.

- Silhouette score of a cluster is taken as the mean of the silhouette scores of its data points.

Silhouette Score Interpretation

- Silhouette score ranges between -1 and 1.



**Poor clusters with potential
wrong assignment of points**

Overlapping clusters

Clearly separated clusters



Power Ahead !

